

Notes on General Relativity and Cosmology

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These notes are an outgrowth of my lecture notes from my general relativity and cosmology course (PHYS 172) taught by Professor Rahulkumar Solanki at Harvey Mudd in Spring 2026. The following additional resources were used to compile these notes.

- *The Classical Theory of Fields* by Landau & Lifshitz. Prof. Solanki's favorite. He followed it quite closely in some sections of the course. Terse introduction to the subject from a legendary series of books.
- *General Relativity* by Wald. Graduate textbook; as Scott Hughes calls it, the "GR überbuch".
- Baumann's notes on cosmology for Part III of Cambridge's mathematical tripos. Recommended to me by Setter. (IN PROGRESS)
- L. Heisenberg's *Review of $f(Q)$ gravity*. Paper that covers quite well the "geometrical trinity of gravity", which refers to three mathematically equivalent classical descriptions of Einstein's general relativity: standard GR (gravity encoded by curvature), teleparallel equivalent of general relativity (gravity encoded by torsion), and symmetric teleparallel equivalent of general relativity (gravity encoded by non-metricity). This is more of an interesting area of active research than part of a standard sequence.

We use the standard Einstein convention (e.g. $a^i b_i \equiv \sum_{i=1}^3 a^i b_i = a^1 b_1 + a^2 b_2 + a^3 b_3$ for $i \in \{1, 2, 3\}$). In the GR part of these notes we use the mostly-plus metric signature and set $G = 1$, and in the cosmology part we use the mostly-minus signature and don't set $G \neq 1$. **Distinguishing GR and cosmology parts and naming the sections properly is something I'll do later.**

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1 Principle of equivalence

- In Newtonian mechanics, the inertial mass is defined by $\vec{F} = m_i \vec{a}$ and the gravitational mass is defined by $\vec{F}_g = -m_g \vec{\nabla} \Phi$, where Φ is the gravitational field.
- The *weak equivalence principle* states that $m_i = m_g$, implying that the behavior of freely falling test particles does not depend on their mass. Consequently, the motion of freely falling particles is the same in a uniform gravitational field and a uniformly accelerated frame.
- The *Einstein equivalence principle* generalizes the statement to all local experiments. It tells us that there are no "gravitationally neutral" objects with respect to which we can measure g . Thus we instead define unaccelerated/inertial frames to be freely falling, which motivates us to describe gravity as not a force, but a property of spacetime.
- Since gravitational fields are inhomogeneous, global inertial frames don't exist.
- Consider an inertial frame K and a uniformly rotating frame of reference K' . In a plane perpendicular to the axis of rotation, the ratio of the circumference to the radius of a circle centered at the origin (through which the axis of rotation passes) in the inertial frame K is 2π . However, a meterstick at rest in K' experiences a Lorentz contraction in the azimuthal direction (viewed from inertial frame K) but not in radial direction since it is perpendicular to the relative velocity (once can consider instantaneous inertial frame of reference coinciding with K' at any instance). Therefore, the ratio of circumference to the radius of the same circle is greater than 2π . Thus, the space is non-Euclidean (or curved) in K' ; however, the spacetime is flat (i.e. Minkowski) because there exists a coordinate transformation through which one can return to inertial frame K . Therefore, from the principle of equivalence, gravitational fields have similar property in a small enough region of spacetime. However, "real" gravitational fields cannot be removed by coordinate transformations, so spacetime must be curved in their presence (which can be made locally flat using appropriate coordinate transformations). Thus, the mathematical theory of gravitational fields is the theory of curved spacetime.
- The propagation of light is also affected by gravitational fields since one cannot distinguish uniform acceleration from gravitational field locally using experiments consisting of light signals. In Newtonian theory, propagation of light has nothing to do with gravity. Furthermore, according to special relativity, the mass of a body is a measure of its energy-content; if the energy changes by l , the mass changes in the same sense by l/c^2 . For example, the mass of the hydrogen atom is less than the sum of masses of proton and electron due to negative binding energy. Therefore, according to the equivalence principle, the gravitational mass has to change in exactly the same manner. Essentially, any form of energy (whether mass or electromagnetic field) produces a gravitational field. Finally, the negative binding energy (gravitational or otherwise) makes the theory of general relativity highly nonlinear, since the gravitational field of the end product is not the vector sum of the field produced by constituents.

2 The geometry of spacetime

2.1 Curvilinear coordinates

- Consider the transformation of coordinates $x^\mu \rightarrow x'^\mu$. The basic transformation is that of the differential:

$$dx^i = \frac{\partial x^\mu}{\partial x'^\nu} dx'^\nu$$

- The derivatives of a scalar ϕ transform as

$$\frac{\partial \phi}{\partial x^i} = \frac{\partial \phi}{\partial x'^k} \frac{\partial x'^k}{\partial x^i}$$

- A *contravariant four-vector* is any aggregate of four quantities A^μ transforming like coordinate differentials under transformation of coordinates. The dual four-vectors are the *covariant four-vectors*. Together, these form the set of *rank-1 tensors* in four-dimensional spacetime¹. The transformation laws are

$$A^\mu = \frac{\partial x^\mu}{\partial x'^\nu} A'^\nu \quad (\text{contravariant vector})$$

$$A_\mu = \frac{\partial x'^\nu}{\partial x^\mu} A'_\nu \quad (\text{covariant vector})$$

- Rank-2 tensors* are aggregates of 16 quantities transforming like

$$A^{\mu\nu} = \frac{\partial x^\mu}{\partial x'^\rho} \frac{\partial x^\nu}{\partial x'^\sigma} A'^{\rho\sigma} \quad (\text{contravariant})$$

$$A_{\mu\nu} = \frac{\partial x'^\rho}{\partial x^\mu} \frac{\partial x'^\sigma}{\partial x^\nu} A'_{\rho\sigma} \quad (\text{covariant})$$

$$A^\mu_\nu = \frac{\partial x^\mu}{\partial x'^\rho} \frac{\partial x'^\sigma}{\partial x^\nu} A'^\rho_\sigma \quad (\text{mixed})$$

- More generally, a tensor of type (r, s) , or rank $r + s$, is an object with r contravariant indices and s covariant indices whose components transform with one Jacobian factor for each index:

$$A^{\mu_1 \dots \mu_r}_{\nu_1 \dots \nu_s} = \frac{\partial x^{\mu_1}}{\partial x'^{\rho_1}} \dots \frac{\partial x^{\mu_r}}{\partial x'^{\rho_r}} \frac{\partial x'^{\sigma_1}}{\partial x^{\nu_1}} \dots \frac{\partial x'^{\sigma_s}}{\partial x^{\nu_s}} A'^{\rho_1 \dots \rho_r}_{\sigma_1 \dots \sigma_s}$$

Thus each upper index transforms like a contravariant vector index, and each lower index transforms like a covariant vector index.

- The *Kronecker delta* δ^μ_ν is a (rank-2) tensor which can "replace indices" like $A^\nu \delta^\mu_\nu = A^\mu$. Note:

$$\frac{\partial x'^\alpha}{\partial x^\mu} \frac{\partial x^\mu}{\partial x'^\beta} = \delta^\alpha_\beta$$

$$\frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x'^\mu}{\partial x^\sigma} = \delta^\rho_\sigma$$

¹From now on, when we speak of tensors, we assume that we are talking about tensors in four-dimensional spacetime, using the physicist's conventions.

- A spacetime is called *flat* if there exists a coordinate system in which the metric takes the form² $g_{\mu\nu} = g^{\mu\nu} = \text{diag}(-1, 1, 1, 1)$ everywhere. Such coordinates are called *galilean coordinates* or *global inertial coordinates*.
- Spacetimes coordinates x'^μ are said to be *curvilinear* if they are related to Galilean coordinates x^μ by a smooth, locally invertible transformation; generally, this transformation is nonlinear, so the coordinate lines need not be straight.
- An infinitesimal *spacetime interval* between two nearby events in curvilinear coordinates is

$$ds^2 = g_{\mu\nu} dx^\mu dx^\nu$$

where $g_{\mu\nu}$ are functions of the spacetime coordinates. $g_{\mu\nu}$ forms a symmetric covariant tensor called the *metric tensor*. The metric tensor raises/lowers indices: $A^\mu = g^{\mu\nu} A_\nu$, $A_\mu = g_{\mu\nu} A^\nu$. This also applies for tensors of higher rank; $A^\mu_\nu = g^{\mu\rho} A_{\rho\nu}$, $A^{\mu\nu} = g^{\mu\rho} g^{\nu\sigma} A_{\rho\sigma}$. In particular, $g_{\alpha\mu} g^{\beta\mu} = \delta^\alpha_\beta$.

- Let x'^μ be galilean coordinates and x^μ arbitrary curvilinear coordinates. Consider the Jacobian of the transformation from the galilean to curvilinear coordinates $J = \frac{\partial(x^0, x^1, x^2, x^3)}{\partial(x'^0, x'^1, x'^2, x'^3)}$. Then the determinant g of the metric tensor $g_{\mu\nu}$ in the system x^μ is given by $J = \frac{1}{\sqrt{-g}}$. So for example, in a galilean coordinate system the integral of a scalar with respect to the four-volume element $d\Omega' = dx'^0 dx'^1 dx'^2 dx'^3$ is also a scalar, so on transforming to curvilinear coordinates, the element of integration goes over into $d\Omega' \rightarrow \sqrt{-g} d\Omega$. Thus, in curvilinear coordinates, when integrating over a four-volume, the quantity $\sqrt{-g} d\Omega$ behaves like an invariant.
- From now on, we denote by τ the *proper time* (i.e. the time experienced in an object's rest frame). For two infinitesimally separated events happening at the same point in space, and denoting by $d\tau$ the (proper) time interval between the two events, we have $ds^2 = -c^2 d\tau^2$, from which $d\tau = \frac{1}{c} \sqrt{-g_{00}} dx^0$. So, the (proper) time between any two events occurring at the same point in space is

$$\tau = \frac{1}{c} \int \sqrt{-g_{00}} dx^0$$

2.2 Covariant differentiation and Christoffel symbols

In galilean coordinates, the differential dA^μ of a vector A^μ forms a vector, and the derivatives $\frac{\partial A^\mu}{\partial x^\nu}$ form a tensor. In general curvilinear coordinates, neither quantity is tensorial, due to the fact that dA^μ is the difference of vectors located at different, infinitesimally separated points of space. In order to obtain a differential of a vector that behaves like a vector. It is necessary that the two vectors to be subtracted from each other be located at the same point in space; that is, we translate one of the vectors to the point where the second is, after which we determine the difference of two vectors which we now refer to one and the same point in space. We also need the operation of translation to be defined so that in galilean coordinates the difference coincides with dA^μ . Under parallel translation of a vector, its components in galilean don't change, but in curvilinear coordinates they generally do (the canonical example is a vector on the equator of a sphere, parallel transported to one of the poles, then back down to the equator in a direction perpendicular to the first displacement, and finally back to the original point on the equator; the

²This is the so called "east coast convention" used in GR. In quantum field theory, and in many treatments of GR, one uses the metric tensor $g_{\mu\nu} = g^{\mu\nu} = \text{diag}(1, -1, -1, -1)$.

final vector comes back rotated by 90 degrees). This ends up being precisely how we do calculus in curvilinear coordinates.

- The change in a vector under a *parallel displacement* is given by

$$DA^\mu = \left(\frac{\partial A^\mu}{\partial x^\rho} + \Gamma_{\nu\rho}^\mu A^\nu \right) dx^\rho$$

$$DA_\mu = \left(\frac{\partial A_\mu}{\partial x^\rho} - \Gamma_{\mu\rho}^\nu A_\nu \right) dx^\rho$$

where the quantities $\Gamma_{\nu\rho}^\mu$ are (local) coefficients called *Christoffel symbols*. The expressions in the parentheses, called the *covariant derivatives* of vectors A^μ , A_μ , denoted $\nabla_\rho A^\mu$ and $\nabla_\rho A_\mu$ respectively, give the desired generalization of the concept of a derivative to curvilinear coordinates.

- One can similarly determine the covariant derivative of a tensor of arbitrary rank; to obtain the covariant derivative of a tensor $A_{\dots}$ with respect to x^ρ , we add to the ordinary derivative $\partial A_{\dots}/\partial x^\rho$ for each covariant index ($A_{\dots\mu}$) a term $-\Gamma_{\mu\rho}^\nu A_{\dots\nu}$, and for each contravariant index ($A^{\dots\mu}$) a term $+\Gamma_{\nu\rho}^\mu A^{\dots\nu}$.
- In a Galilean coordinate system, the Christoffel symbols vanish, and covariant differentiation reduces to ordinary differentiation.
- In terms of the metric tensor,

$$\Gamma_{\nu\rho}^\mu = \frac{g^{\mu\sigma}}{2} (\partial_\nu g_{\rho\sigma} + \partial_\rho g_{\nu\sigma} - \partial_\sigma g_{\nu\rho}) = \Gamma_{\rho\nu}^\mu$$

- The Christoffel symbols are not tensors: they behave like a tensor only under linear transformations. Generally, they transform as

$$\Gamma_{\mu\lambda}^{\nu'} = \frac{\partial x^{\nu'}}{\partial x^\alpha} \frac{\partial x^\rho}{\partial x'^\mu} \frac{\partial x^\sigma}{\partial x'^\lambda} \Gamma_{\rho\sigma}^\alpha - \frac{\partial x^\alpha}{\partial x'^\mu} \frac{\partial x^\beta}{\partial x'^\lambda} \frac{\partial^2 x^{\nu'}}{\partial x^\alpha \partial x^\beta}$$

- The covariant derivative as defined above indeed follows the conditions one requires of a derivative operator: linearity, the Leibnitz (product) rule, commutativity with contraction, consistency with the notion of tangent vectors as directional derivatives on scalar fields, and finally, the "torsion-free" condition $\nabla_\mu \nabla_\nu f = \nabla_\nu \nabla_\mu f$ for scalar f . While this last condition is sometimes dropped, in GR the derivative operator is assumed to be torsion-free.
- The metric tensor acts as a constant under covariant differentiation, that is, $\nabla_\rho g_{\mu\nu} = 0$; in fact, for any given metric, the derivative operator ∇_ρ satisfying this condition exists and is unique.
- If in a covariant derivative we raise the index signifying the differentiation, we obtain the so-called *contravariant derivative*; thus, $\nabla^\nu A_\mu = g^{\nu\rho} \nabla_\rho A_\mu$, $\nabla^\nu A^\mu = g^{\nu\rho} \nabla_\rho A^\mu$.
- It is always possible to choose a coordinate system in which all the $\Gamma_{\nu\rho}^\mu$ become zero at a previously assigned point. Such a system is said to be *locally inertial*. In fact, let a given point be chosen as the origin of coordinates, and let the values of $\Gamma_{\nu\rho}^\mu$ at it be initially (in the coordinates x^μ) equal to $(\Gamma_{\nu\rho}^\mu)_0$. In the neighborhood of this point, make the transformation $x'^\mu = x^\mu + \frac{1}{2} (\Gamma_{\nu\rho}^\mu)_0 x^\nu x^\rho$. Then all the $\Gamma_{\nu\rho}^\mu$ become zero. Such a transformation has $\left(\frac{\partial x'^\mu}{\partial x^\nu} \right)_0 = \delta_\nu^\mu$, so it does not change the value of any tensor at the given point, so we can make the Christoffel symbols vanish at the same time as we bring the $g_{\mu\nu}$ to Cartesian form.

- A vector A^μ given at each point on some curve C with a tangent B^μ is said to be *parallelly transported* as one moves along the curve if $B^\mu \nabla_\mu A^\nu = 0$ along the curve. More generally, one can define the parallel transport of a tensor of arbitrary rank by $B^\mu \nabla_\mu A_{\sigma_1 \dots \sigma_s}^{\rho_1 \dots \rho_r}$. Choosing a coordinate basis and parametrizing C by t , $\frac{dA^\nu}{dt} + B^\mu \Gamma_{\mu\lambda}^\nu A^\lambda = 0$.

2.3 Curvature

Euclid's fifth postulate states that if two lines are drawn which intersect a third in such a way that the sum of the inner angles on one side is less than two right angles, then the two lines inevitably must intersect each other on that side if extended far enough. Two-dimensional space of constant negative curvature satisfies all but this postulate of Euclid. There are two approaches to studying the curvature of an object: the *extrinsic approach*, which is to analyze the object by embedding it in a higher-dimensional space, or to analyze it *intrinsically* by measuring entirely from within, without referencing any higher-dimensional surrounding space. General relativity studies curved spacetime intrinsically.

- Let $\delta A_\mu = \Gamma_{\mu\nu}^\gamma A_\gamma dx^\nu$ be the change in DA_μ under the parallel transport from $x^\mu \rightarrow x^\mu + dx^\mu$, i.e. $DA_\mu = dA_\mu + \delta A_\mu$. This δA_μ is path-dependent due to the dx^ν term, and therefore so is the covariant difference. The change in an arbitrary vector field after parallel displacement around any infinitesimal closed contour gives us necessary and sufficient conditions for whether or not spacetime is curved. Write this change $\Delta A_\mu = \oint \delta A_\mu = \oint \Gamma_{\mu\nu}^\lambda A_\lambda dx^\nu$. Then, with the first-order accuracy sufficient for the transformation, $\Delta A_\mu = \frac{1}{2} R_{\mu\nu\sigma}^\lambda A_\lambda \Delta f^{\nu\sigma}$, where $\Delta f^{\nu\sigma}$ is the infinitesimal value of the area enclosed by the contour, and $R_{\mu\nu\sigma}^\lambda$ is the curvature tensor defined below.
- Define the *curvature tensor* (or *Riemann tensor*)

$$R_{\mu\nu\alpha\beta} A^\nu = (\nabla_\alpha \nabla_\beta - \nabla_\beta \nabla_\alpha) A_\mu$$

Equivalently,

$$\begin{aligned} R_{\nu\rho\sigma}^\mu &= \partial_\rho \Gamma_{\nu\sigma}^\mu - \partial_\sigma \Gamma_{\nu\rho}^\mu + \Gamma_{\alpha\rho}^\mu \Gamma_{\nu\sigma}^\alpha - \Gamma_{\alpha\sigma}^\mu \Gamma_{\nu\rho}^\alpha \\ R_{\mu\nu\rho\sigma} &= \frac{1}{2} (\partial_\nu \partial_\rho g_{\mu\sigma} + \partial_\mu \partial_\sigma g_{\nu\rho} - \partial_\nu \partial_\sigma g_{\mu\rho} - \partial_\mu \partial_\rho g_{\nu\sigma}) + g_{\alpha\beta} (\Gamma_{\nu\rho}^\alpha \Gamma_{\mu\sigma}^\beta - \Gamma_{\nu\sigma}^\alpha \Gamma_{\mu\rho}^\beta) \end{aligned}$$

This tensor vanishes if and only if the curvature is zero. By way of coordinate transformations, we can make $\Gamma_{\beta\sigma}^\alpha = 0$ at a single point, but $\partial_\beta \Gamma_{\beta\gamma}^\alpha \neq 0$ if the spacetime is curved.

- Properties of the Riemann tensor include:

$$\begin{aligned} R_{\alpha\beta\gamma\delta} &= -R_{\beta\alpha\gamma\delta} = R_{\beta\alpha\delta\gamma} = R_{\gamma\delta\alpha\beta} g^{\mu\nu} R_{\mu\nu\rho\sigma} = R_{\mu\rho\sigma}^\mu = 0 \\ R_{\alpha\beta\gamma\delta} + R_{\beta\gamma\alpha\delta} + R_{\gamma\alpha\beta\delta} &= 0 \\ \nabla_\alpha R_{\beta\gamma\delta\sigma} + \nabla_\beta R_{\gamma\alpha\delta\sigma} + \nabla_\gamma R_{\alpha\beta\delta\sigma} &= 0 \end{aligned}$$

- The trace of the curvature (or Riemann) tensor is captured by the *Ricci tensor* $R_{\nu\sigma} = g^{\mu\rho} R_{\mu\nu\rho\sigma} = R_{\nu\mu\sigma}^\mu$ and tracefree part is captured by the *Weyl tensor* $C_{\alpha\beta\gamma\delta} = R_{\alpha\beta\gamma\delta} - \frac{1}{2} R_{\alpha\gamma} g_{\beta\delta} + \frac{1}{2} R_{\alpha\delta} g_{\beta\gamma} + \frac{1}{2} R_{\beta\gamma} g_{\alpha\delta} - \frac{1}{2} R_{\beta\delta} g_{\alpha\gamma} + \frac{R}{6} (g_{\alpha\gamma} g_{\beta\delta} - g_{\alpha\delta} g_{\beta\gamma})$. In some notations, one writes $C_{\rho\sigma}^{\mu\nu} = R^{\mu\nu}{}_{\rho\sigma} - 2\delta_{[\rho}^{[\mu} R_{\sigma]}^{\nu]} + \frac{R}{3} \delta_{[\rho}^{[\mu} \delta_{\sigma]}^{\nu]}$, where the square bracket around indices represents antisymmetrization $A_{[\alpha\beta]} = \frac{1}{2} (A_{\alpha\beta} - A_{\beta\alpha})$.
- The *Ricci scalar* is $g^{\nu\sigma} R_{\nu\sigma} = R$.

2.4 Geodesics

- The equations of motion for a free motion of a particle in special relativity are $\frac{du^\mu}{d\tau} = 0$ or $du^\mu = 0$, where $u^\mu = \frac{dx^\mu}{d\tau}$ is the *four-velocity*. It follows by definition that $u^\mu u_\mu = -1$.
- We postulate that the motion of freely falling massive particles (i.e. particles traveling with speed less than c under the influence of only gravity) is given by the path of maximum proper time, called a *timelike geodesic*. Physically, this postulate makes sense because such paths are locally straight through spacetime, just like segments of minimum length are straight lines in Euclidean space. A physical approach to finding this motion would be to set $a^\mu = \frac{Du^\mu}{D\tau} = 0$. Equivalently, such particles satisfy the equations of motion

$$\frac{du^\mu}{d\tau} + \Gamma_{\nu\rho}^\mu u^\nu u^\rho = \frac{d^2 x^\mu}{d\tau^2} + \Gamma_{\nu\rho}^\mu \frac{dx^\nu}{d\tau} \frac{dx^\rho}{d\tau} = 0$$

We call this the *timelike geodesic equation*.

- Alternatively, we can define a timelike geodesic to be a timelike curve $x^\mu = x^\mu(\sigma)$ joining two events A and B , where σ is some parameter, satisfying

$$\delta S = \delta \left(\int_A^B \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma}} d\sigma \right) = 0$$

among all the timelike curves in the vicinity joining A and B . Recall that the tangent to such a timelike curve is $\frac{dx^\mu}{d\sigma}$. Curves satisfying the above then follow the Euler-Lagrange equations $\frac{\partial \mathcal{L}}{\partial x^\alpha} - \frac{d}{d\sigma} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\alpha} \right) = 0$ where $L = \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma}}$; then the first definition is recovered.

- A spacelike path cannot be travelled by massive or massless particles since the events lying on these paths are non-causal. However, spacelike geodesics can be useful in defining spatial coordinates, for example, in the case of Fermi normal coordinates. If σ is a parameter for a spacelike curve $x^\mu = x^\mu(\sigma)$, then $g_{\mu\nu} \frac{dx^\mu}{d\sigma} \frac{dx^\nu}{d\sigma} > 0$ and one can proceed in a similar manner as the previous parts to find a geodesic equation with proper length s (instead of τ):

$$\frac{d^2 x^\mu}{ds^2} + \Gamma_{\nu\rho}^\mu \frac{dx^\nu}{ds} \frac{dx^\rho}{ds} = 0$$

- For a null curve, $ds^2 = 0$, so it does not make sense to apply the variational principle since all null paths will give total spacetime interval between fixed events A, B to equal zero. If $x^\mu = x^\mu(\lambda)$ is a null curve, then one generally chooses λ such that the wave four-vector is $k^\mu = \frac{dx^\mu}{d\lambda}$. Since we consider geodesics as generalizations of straight lines, $Dk^\mu = 0$ is the geodesic equation. We have thus obtained the *lightlike geodesic equation*

$$\frac{dk^\mu}{d\lambda} + \Gamma_{\nu\sigma}^\mu k^\nu k^\sigma = 0 = \frac{d^2 x^\mu}{d\lambda^2} + \Gamma_{\nu\sigma}^\mu \frac{dx^\nu}{d\lambda} \frac{dx^\sigma}{d\lambda}$$

- It is very important to understand the difference between a null path and a null geodesic; for a null path, $k^\mu k_\mu = 0$, and null geodesics satisfy the geodesic equation in addition to the null path condition.

3 Einstein's general relativity

3.1 The energy-momentum tensor

- The *(energy-)momentum four-vector* p^μ of a particle of mass m is defined by $p^\mu = mu^\mu$. The *energy* of a particle as measured by an observer present at the site of the particle whose 4-velocity is v^μ is defined by $E = -p_\mu v^\mu$. Thus, in special relativity, energy is recognized to be the time component of p^μ . For a particle at rest with respect to the observer (i.e. $v^\mu = u^\mu$), this reduces to the familiar $E = mc^2$. From now on, we use natural units, setting $c = 1$.
- We find that $p^\mu = mu^\mu$, $p^\mu p_\mu = -m^2$.
- Continuous matter distributions in special relativity are described by a symmetric (*stress-energy-momentum*) tensor.
- Consider the Lorentz transformation

$$\begin{aligned} x'^0 &= \frac{x^0 - ux^1}{\sqrt{1-u^2}} & x'^1 &= \frac{x^1 - ux^0}{\sqrt{1-u^2}} \\ x'^2 &= x^2 & x'^3 &= x^3 \end{aligned}$$

moving from a frame S to a frame S' moving at a velocity u in the $+x$ direction relative to S . The line element $d\ell^2 = -dt^2 + dx^2 + dy^2 + dz^2$ is invariant under such transformations.

- Under such transformations, a four-vector A^μ transforms as

$$\begin{aligned} A'^0 &= \frac{A^0 - uA^1}{\sqrt{1-u^2}} & A'^1 &= \frac{A^1 - uA^0}{\sqrt{1-u^2}} \\ A'^2 &= A^2 & A'^3 &= A^3 \end{aligned}$$

- Unlike mass, density is not an invariant, so it transforms under Lorentz transformations. For the treatment of charge density (which is equivalent to that of number density), consider the charge enclosed by some volume V in S . When we Lorentz-transform, the charge in the corresponding volume V' in S' is equal to the charge enclosed by V . Thus $\rho dV = \rho' dV' = \rho_0 dV_0$, where the last quantity refers to the proper frame, i.e. where the charges are at rest. Let v, v' denote the velocities of S, S' w.r.t. the proper frame, respectively. Then:

$$\begin{aligned} dV &= dV_0 \sqrt{1-v^2} & \rho &= \frac{\rho_0}{\sqrt{1-v^2}} \\ dV' &= dV_0 \sqrt{1-v'^2} & \rho' &= \frac{\rho_0}{\sqrt{1-v'^2}} \end{aligned}$$

yielding $\rho' = \rho \sqrt{\frac{1-v^2}{1-v'^2}}$. Recalling velocity transformations $v' = \frac{v-u}{1-uv}$,

$$\rho' = \frac{\rho(1-uv)}{\sqrt{1-u^2}} = \frac{\rho - u(\rho v)}{\sqrt{1-v^2}}$$

- $\vec{J} = \rho \vec{v}$ is the *charge flux* or *current density*, i.e. the charges per unit area per unit time.
- Now consider a case in which the charges are in S , moving in a direction normal to some fixed cross sectional area A at a velocity v . In time dt , the amount of charges passing through A is given by $\rho A v dt$, so the charge flux is ρv . Now, $\rho' v' = \frac{\rho(1-uv)}{\sqrt{1-u^2}} \left(\frac{v-u}{1-uv} \right) = \frac{\rho(v-u)}{\sqrt{1-u^2}}$.

- We construct the *four-current density*

$$J^\mu = (\rho, \vec{J}) = (\rho, \rho \vec{v}) = \rho_0 \left(\frac{1}{\sqrt{1-v^2}}, \frac{\vec{v}}{\sqrt{1-v^2}} \right) = \rho_0 u^\mu$$

- A *global conservation* law for some quantity (e.g. for charge or energy) tells us that the integral of that quantity's density over all volume is constant. Then *local conservation* needs to hold for causality; this yields a *continuity* condition that is, for some arbitrary volume V enclosed by a surface S , local conservation requires that the rate of change of total charges inside V equal the number of charges leaving or entering surface S per unit time. Mathematically,

$$\begin{aligned} -\frac{\partial}{\partial t} \iiint_V \rho dV &= \oint_S \vec{J} \cdot d\vec{S} \\ \Rightarrow \iiint_V -\frac{\partial \rho}{\partial t} dV &= \iiint_V (\nabla \cdot \vec{J}) dV \\ \Rightarrow \frac{\partial S}{\partial t} + \nabla \cdot \vec{J} &= 0 \end{aligned}$$

In terms of four-vectors, we can write $\frac{\partial J^\mu}{\partial x^\mu} = 0$. The covariant version is simply $\nabla_\mu J^\mu = 0$.

- For energy density, we have the same setup. Because energy is a component of the four-momentum, which is a four-vector, it transforms under a Lorentz transform accordingly:

$$E' = \frac{E - up_x}{\sqrt{1-u^2}}, \quad p'_x = \frac{p_x - uE}{\sqrt{1-u^2}}$$

- Energy flux in the $+x$ direction is the amount of energy passing through unit area in yz plane in unit time. Thus energy flux equals energy density $\frac{\rho_0}{\sqrt{1-v^2}}$ times velocity; this is just then momentum density. This gives us the *energy-momentum tensor for pressureless dust*:

$$T^{\mu\nu} = \rho u^\mu u^\nu$$

- For a particle, a rank 1 four-momentum tells us how energy and momentum transform from one frame to another; for a continuous distribution of matter, a rank 2 energy-momentum tensor tells us how energy and momentum densities transform from one frame to another.
- T^{00} is the energy density, T^{i0} is the (3-)momentum density, T^{0i} is the energy flux, and T^{ik} is the stress tensor or momentum flux. $T^{\mu\nu} = T^{\nu\mu}$ due to energy flux equaling momentum density and the stress tensor being symmetric (due to conservation of angular momentum).
- In terms of $T^{\mu\nu}$, conservation of energy is $\nabla_\mu T^{0\mu} = 0$, and conservation of momentum (in the i th direction) is $\nabla_\mu T^{i\mu} = 0$. Thus, in general, $\nabla_\mu T^{\mu\nu} = 0$.
- Instead of pressureless dust, consider a perfect fluid with isotropic pressure. In the rest/proper/comoving frame of the fluid, there is only isotropic pressure P and no dissipation or energy flux. Therefore, for a perfect fluid in the rest frame,

$$T^{\mu\nu} = \begin{pmatrix} \rho & 0 & 0 & 0 \\ 0 & P & 0 & 0 \\ 0 & 0 & P & 0 \\ 0 & 0 & 0 & P \end{pmatrix}$$

In general,

$$T^{\mu\nu} = (\rho + P)u^\mu u^\nu + Pg^{\mu\nu}$$

The electromagnetic field behaves somewhat like a pressurized fluid; it turns out that for an electromagnetic field $T^{\mu\nu} = F^{\mu\alpha}F^\nu_\alpha - \frac{g^{\mu\nu}}{4}F_{\alpha\beta}F^{\alpha\beta}$ where $F^{\mu\nu}$ is the electromagnetic tensor.

- The most general form of the energy-momentum tensor is

$$T^{\mu\nu} = (\rho + P)u^\mu u^\nu + q^\mu u^\nu + q^\nu u^\mu + Pg^{\mu\nu} + \pi^{\mu\nu}$$

where q^μ is energy flux and $\pi^{\mu\nu}$ is (traceless) anisotropic pressure.

3.2 Einstein's field equations

- Matter tells spacetime how to curve and spacetime tells matter how to move. The curvature of spacetime should then correspond to the energy-momentum of matter/fluid. But the latter is (4-)divergence-free, i.e. $\nabla_\mu T^{\mu\nu} = 0$, whereas the rank-2 tensor we use to describe curvature—the Ricci tensor—is not divergence-free. Using the Bianchi identity, one defines a divergence-free tensor, called the *Einstein tensor*:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{R}{2}g_{\mu\nu}$$

- *Einstein's field equations* are

$$G_{\mu\nu} = 8\pi T_{\mu\nu}$$

- The formal classical field theory approach to derive this is to use the *Einstein-Hilbert* action

$$S = S_{\text{gravity}} + S_{\text{matter}} = \int R\sqrt{-g}d\Omega + \int T_{\mu\nu}\delta g^{\mu\nu}\sqrt{-g}d\Omega$$

where $g = \det(g_{\mu\nu})$ and $d\Omega$ is a four-volume element. This can be compared to classical electrodynamics, where

$$S = S_{\text{field}} + S_{\text{interactions}} + S_{\text{matter}} = \int F_{\mu\nu}F^{\mu\nu}\sqrt{-g}d\Omega + \int A_\mu J^\mu\sqrt{-g}d\Omega + \int \mu\sqrt{-g}d\Omega$$

where μ is mass density. The equation $\delta S_{\text{interactions}} + \delta S_{\text{matter}} = 0$ gives the Lorentz force $F^{\alpha\beta}J_\beta = \mu a^\alpha$, and $\delta S_{\text{field}} + \delta S_{\text{interactions}} = 0$ gives the field equation $\nabla_\mu F^{\mu\nu} = -J^\nu$.

- Rank 2 tensors have 16 components. Symmetric rank 2 tensors thus have 10 independent components. Conservation of energy $\nabla_\mu T^{\mu\nu} = 0$ means there are 4 constraints or 4 degrees of freedom that can be gauge fixed. Thus, there are 6 "independent" field equations.

3.3 Schwarzschild

- The first *exact* solution of Einstein's field equation was found by Schwarzschild. He gave two solutions, interior and exterior, where the former refers to the gravitational field inside a spherically symmetric constant density star, whereas the latter refers to the gravitational field outside any spherically symmetric star, that is, a vacuum solution.
- In vacuum, $T_{\mu\nu} = 0 = R_{\mu\nu}$, thus $R = 0$.

- A solution is called *asymptotically flat* if, infinitely far away from the spherical body producing the gravitational field, the spacetime becomes flat/Minkowski.
- A solution is called *static* if spacetime does not change over time and is also irrotational. This is a spatial case of *stationary spacetime*, in which the geometry does not change in time but can rotate.
- *Birkhoff's theorem* states that any spherically symmetric solution of the vacuum field equations (e.g. the Schwarzschild exterior solution, which is unique) is static and asymptotically flat.
- The solution being static means that spherically symmetric gravitational waves don't exist, even though matter in spherical stars is allowed to move/oscillate radially.
- We consider the static solution. A static solution has a line element of the form

$$ds^2 = g_{tt}dt^2 + g_{ik}dx^i dx^k$$

where $\partial_t g_{\mu\nu} = 0$.

- The most general spherically symmetric static line element is

$$ds^2 = -e^{2v(r)}dt^2 + e^{2\lambda(r)}dr^2 + r^2(d\theta^2 + \sin^2\theta d\phi^2)$$

The Einstein tensor is then

$$G_t^t = -\frac{e^{2\lambda}}{r^2}(2r\lambda' + e^{2\lambda} - 1) \quad G_r^r = -\frac{e^{2\lambda}}{r^2}(e^{2\lambda} - 1 - 2rv')$$

$G_\theta^\theta = G_\phi^\phi \neq 0$ but we don't need it to find $\lambda(r), v(r)$. Now, $G_t^t = 0 \implies \frac{d}{dr}(re^{-2\lambda}) = 1 \implies e^{-2\lambda} = (1 - \frac{r_s}{r})$, where r is an integration constant; furthermore, $G_r^r = 0$ and $e^{2v} = (1 - \frac{r_s}{r})$. We take $r_s = 2M$ for reasons which will be clear in the next section. Basically, in the weak field limit, M is mass or newtonian mass.

- Thus the exterior Schwarzschild solution reduces to

$$ds^2 = -\left(1 - \frac{2M}{r}\right)dt^2 + \frac{dr^2}{\left(1 - \frac{2M}{r}\right)} + r^2(d\theta^2 + \sin^2\theta d\phi^2)$$

If one inserts G and c back then $|g_{tt}| = 1 - \frac{2GM}{rc^2}$.

3.4 The Newtonian limit

- Consider a spherically symmetric gravitational field produced by a star in Newtonian gravity. The equation of motion of a test particle in a radial ingoing free fall is given by $\vec{a} = -\nabla\Phi \implies \frac{d^2r}{dt^2} = \frac{M}{r^2}$ where we take $G = 1$.
- The same notion in GR is the geodesic equation with $\frac{d\theta}{d\tau} = \frac{d\phi}{d\tau} = 0$. We consider equatorial

plane without loss of generality i.e. $\theta = \pi/2$. One computes the Christoffel symbols

$$\begin{aligned}\Gamma_{tt}^r &= \frac{M}{r^2} \left(1 - \frac{2M}{r}\right) & \Gamma_{rt}^r &= \frac{-M}{r^2} \left(1 - \frac{2M}{r}\right) = -\Gamma_{tr}^t \\ \Gamma_{r\theta}^\theta &= \frac{1}{r} = \Gamma_{r\phi}^\phi & \Gamma_{\theta\theta}^r &= -r \left(1 - \frac{2M}{r}\right) \\ \Gamma_{\phi\phi}^r &= -r \sin^2 \theta \left(1 - \frac{2M}{r}\right) & \Gamma_{\phi\phi}^\theta &= -\sin \theta \cos \theta \\ & & \Gamma_{\theta\phi}^\phi &= \cot \theta\end{aligned}$$

- To compare Newtonian and general relativistic trajectories, far from the body producing gravitational field, coordinate length and time become proper length and time due to asymptotic flatness; in otherwords, we need to compare $\frac{d^2 r}{dt^2}$ for GR using geodesic equation where r is quite large. Using

$$\frac{d^2 r}{d\tau^2} = \frac{d^2 r}{dt^2} \dot{t}^2 + \frac{dr}{dt} \ddot{t}$$

the geodesic equations for r and t become

$$\begin{aligned}\ddot{r} + \Gamma_{rr}^r \dot{r}^2 + \Gamma_{tt}^r \dot{t}^2 &= 0 \\ \ddot{t} + \Gamma_{tr}^t \dot{r} \dot{t} &= 0\end{aligned}$$

Plugging and simplifying and reinserting G, c ,

$$\frac{d^2 r}{dt^2} = -\frac{GM}{r^2} + \frac{2G^2 M^2}{r^3 c^2}$$

so in the newtonian limit $t \rightarrow \tau$ and $c \rightarrow \infty$ we recover regular equation of motion for Newtonian gravity. Thus, the parameter M in the Schwarzschild solution is mass in the Newtonian limit.

3.5 Newtonian orbits

In the following discussion, we work in polar coordinates. By conservation of energy, for a Newtonian orbit $E_n = K + V = \frac{1}{2}mv^2 - \frac{GMm}{r}$ is conserved. The equation of motion in r , with $\dot{\theta}$ expressed in terms of conserved angular momentum $l = mr^2\dot{\theta}$, involves only r and its derivatives. It is the same equation as would be obtained for a fictitious one-dimensional problem in which a particle of mass m is subject to a force $f' = f + \frac{l^2}{mr^3}$. The significance of the additional term is clear if it is written as $mr\dot{\theta}^2 = \frac{mv_\theta^2}{r}$, which is the familiar centrifugal force. Then the equation of motion of the particle in r is that of a one-dimensional problem with a fictitious potential $V' = V + \frac{1}{2} \frac{l^2}{mr^2}$. So the effective potential energy is $V = -\frac{GMm}{r} + \frac{l^2}{2mr^2}$; plotting this against r gives a potential well. At the bottom of this potential well are circular orbits. At energies where a horizontal line intersects the plotted curve at 2 points, we have elliptic orbits. If it intersects at only one point, we have parabolic/hyperbolic orbits; unless they are directly headed towards each other, the masses pass each other, never to meet again.

3.6 Measurements

- The energy of a particle measured by the observer moving with four-velocity u^μ is $E = -p_\mu u^\mu$.
- For massless particles, since our wave four-vector k^μ is related to p^μ by some constant, we can replace $p^\mu \rightarrow k^\mu$ and $E \rightarrow \omega$, yielding $\omega_0 = -k_\mu u^\mu$.

3.7 Schwarzschild geodesics

- In the Schwarzschild metric, $L = \sqrt{-g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau}} = \sqrt{\left| -f(r)\dot{t}^2 + \frac{\dot{r}^2}{f(r)} + r^2\dot{\phi}^2 \right|}$ where $\dot{t} = \frac{dt}{d\lambda}$ and $f(r) = \left(1 - \frac{2M}{r}\right)$.
- One can take a shortcut by using the Lagrangian $\mathcal{L} = -f(r)\dot{t}^2 + \frac{\dot{r}^2}{f(r)} + r^2\dot{\phi}^2$ where we parametrize directly using proper time i.e. $\dot{t} = \frac{dt}{d\tau}$. Take $\mathcal{L} = \epsilon$ where ϵ is a constant of motion, 0 for null geodesics and -1 for timelike geodesics, and the geodesic equation is $\frac{\partial \mathcal{L}}{\partial x^\mu} = \frac{d}{d\lambda} \left(\frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \right)$.
- Generalized momenta are defined similarly to in the Lagrangian approach. For timelike geodesics, $\frac{p_\mu}{m} = \frac{\partial \mathcal{L}}{\partial \dot{x}^\mu} \Big|_{L=1, \tau=\lambda}$ where m is the mass of the particle.
- In general, we normalize four-velocity so $L^2 = -1$ and $L = 1$.
- Since $\frac{\partial \mathcal{L}}{\partial t} = 0$, energy per unit rest mass is conserved, that is, $\frac{p_t}{m} = -E = -f(r)\frac{dt}{d\tau} = -f(r)\dot{t} = \text{constant}$.
- Similarly, since $\frac{\partial \mathcal{L}}{\partial \phi} = 0$, momentum per unit mass is conserved. That is, $\frac{p_\phi}{m} = J = r^2\dot{\phi} = \text{constant}$.
- So the conserved quantities are $\mathcal{L}$, E , and J .
- From the constants of motion,

$$\frac{E^2 - 1}{2} = \frac{1}{2} \left(\frac{dr}{d\tau} \right)^2 - \frac{M}{2} + \frac{J^2}{2r^2} - \frac{MJ^2}{r^3}$$

where effective potential

$$V_{eff} = -\frac{M}{r} + \frac{J^2}{2r^2} - \frac{MJ^2}{r^3} = -\left(\frac{r}{M}\right)^{-1} + \frac{1}{2} \left(\frac{J/M}{r/M}\right)^2 - \frac{(J/m)^2}{(r/M)^3}$$

To compare with Newtonian orbits let's reinstate c, G and write E as the energy per unit mass $\frac{E_n + mc^2}{mc^2}$ and J as the angular momentum per unit mass J_n/M . To take the Newtonian limit $c \rightarrow \infty$ we recover our usual Newtonian orbit expression ($\tau \rightarrow t$)

$$E_n = \frac{m}{2} \left(\frac{dr}{dt} \right)^2 + \frac{J_n^2}{2mr^2} - \frac{GMm}{r}$$

where the last two terms constitute the effective potential.

- $\frac{dV_{eff}}{dr} = 0$ when $r_{\pm} = \frac{J^2}{2M} \left[1 \pm \sqrt{1 - 12 \left(\frac{M}{J} \right)^2} \right]$; $r_+ = r_{min}$ is where stable circular orbit occurs. (to add: V_{eff} vs. r/M plot, LN11-13P8) Thus, the location of circular orbit depends upon $\frac{J}{M}$. If $\frac{M}{J} > \sqrt{12}$ or $\frac{J}{M} < \sqrt{12}$, we don't have equilibrium points. Moreover, $\frac{J}{M} = \sqrt{12}$ (i.e. $r = 6M$) is the location of the inner most stable circular orbit (ISCO).
- For a timelike geodesic $\mathcal{L} = -1$, rearranging variables gives

$$\left(\frac{dr}{d\phi} \right)^2 = \frac{E^2 r^4}{J^2} - r^2 \left(1 - \frac{2M}{r} \right) \left(\frac{r^2}{J^2} + 1 \right)$$

Integrating gives the precession

$$\Delta\phi = -2r + 2 \int_{r_{min}}^{r_{max}} \frac{J dr}{r^2 \sqrt{E^2 - \left(1 - \frac{2M}{r}\right) \left(1 + \frac{J^2}{r^2}\right)}}$$

As can be seen from the figure, $\frac{dr}{d\phi} = 0$ at $r_{max} = r_+$ and $r_{min} = r_-$, thus $\frac{E^2}{J^2} = f(r_{\pm}) \left(\frac{1}{J^2} + \frac{1}{r_{\pm}^2} \right)$.

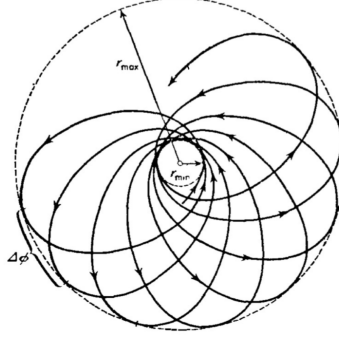


Figure 1: From Landau & Lifshitz Classical Mechanics.

After applying Newtonian approximations, one finds

$$\Delta\phi \approx \frac{6\pi GM}{2} \cdot \left(\frac{1}{r_+} + \frac{1}{r_-} \right) \text{ rad/revolution}$$

- For Mercury, $\frac{1}{2} \left(\frac{1}{r_+} + \frac{1}{r_-} \right) = 55.3 \times 10^6 \text{ km}$ and $M_{\odot}G = 1.475 \text{ km}$ gives $\Delta\phi = 0.1038''$ per revolution. Since Mercury makes 415 revolutions per century, $\Delta\phi = 43.03''$ per century.³
- For a null geodesic $\mathcal{L} = 0$, we can write

$$\frac{1}{J^2} \left(\frac{dr}{d\lambda} \right)^2 + W_{eff}(r) = \frac{E^2}{J^2} = \frac{1}{b^2}$$

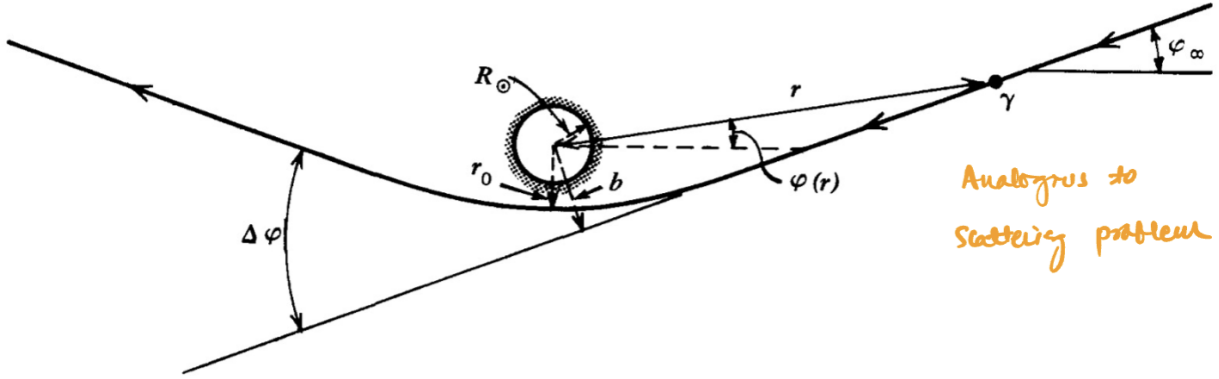
where $\dot{t} = \frac{dt}{d\lambda}$ and $W_{eff} = \frac{1}{r^2} \left(1 - \frac{2M}{r} \right)$. Here the constants of motion $E = f(r)\dot{t}$ and $J = r^2\dot{\phi}$ are energy and angular momentum of the particle following the null geodesic. Basically, $k_{\mu} = \frac{\partial \mathcal{L}}{\partial x^{\mu}}$, so $k_t = -E \equiv -\omega = -f(r)\frac{dt}{d\lambda}$ and $k_{\phi} = J = r^2\frac{d\phi}{d\lambda}$ are conserved. Here, $k_{\mu}k^{\mu} = 0$ and k_{μ} is the wave four-vector (or four-momentum).

- Unlike timelike geodesics where only proper time yields $\mathcal{L} = -1$, any affine parameter λ that is multiplied by a constant gives $\mathcal{L} = 0$ for null geodesics. Thus, only the ratio of E and J determines the shape of the orbit.
- For a null geodesic, rearranging the expression for $\mathcal{L} = 0$ gives

$$\left(\frac{dr}{d\phi} \right)^2 + r^2 - 2Mr - \frac{E^2}{J^2} r^4 = 0$$

At the closest approach $r = r_0$, $\frac{dr}{d\phi} = 0$, so

³See Weinberg's *Gravitation and Cosmology*.



$$\frac{E^2}{J^2} = \frac{1}{r_0^2} \left(1 - \frac{2M}{r_0} \right)$$

$$\phi = \int \frac{dr/r^2}{\sqrt{f(r_0)/r_0^2 - f(r)/r^2}}$$

By symmetry, $\Delta\phi = \pi - 2|\phi_{r_0} - \phi_\infty|$. For a star, $r_0 \gg 2M = \frac{2GM}{c^2}$, e.g. for the sun $\frac{2GM_\odot}{c^2} \sim 3$ km whereas $R_\odot = 7 \cdot 10^5$ km. Thus, we can take $\frac{2M}{r} \ll 1$ to apply Taylor expansion for approximation. This yields $\delta\phi \approx \frac{4M}{r_0} \approx \frac{4M}{b}$ where b is the impact parameter.

3.8 Gravitational redshift

- Consider an observer moving with unit four-velocity u_o^μ and emitter with u_e^μ . The wave four-vector (or four-momentum) of light follows the null geodesic equation $\frac{dk^\mu}{d\lambda} + \Gamma_{\nu\sigma}^\mu k^\nu k^\sigma = 0$ with $k^\mu k_\mu = 0$. The frequency at which the emitter transmits light is $\omega_e = -k_\mu u_e^\mu$, and the observer receives light at $\omega_o = -k_\mu u_o^\mu$. The redshift in wavelength then is

$$z = \frac{\lambda_o - \lambda_e}{\lambda_e} = \left(\frac{\lambda_o}{\lambda_e} \right) - 1 = \left(\frac{\omega_e}{\omega_o} \right) - 1$$

$$= \left(\frac{k_\mu u_e^\mu}{k_\mu u_o^\mu} \right) - 1$$

- For a stationary observer and emitter in Schwarzschild spacetime, $u_o^k = u_e^k = 0$; from the line element, $-d\tau^2 = g_{tt}dt^2$, so $u_o^0 = \frac{1}{\sqrt{-g_{tt}(o)}}$ and $u_e^0 = \frac{1}{\sqrt{-g_{tt}(e)}}$. Then $-k_\mu u^\mu|_o = \omega_o = \frac{\omega}{\sqrt{-g_{tt}(o)}}$ and $\omega_e = \frac{\omega}{\sqrt{-g_{tt}(e)}}$, and since $\omega = E = \frac{1}{2} \frac{\partial \mathcal{L}}{\partial t} = -\left(1 - \frac{2M}{r}\right) \dot{t}$ is constant, the redshift is

$$z = \left(\frac{1 - \frac{2M}{r_o}}{1 - \frac{2M}{r_e}} \right)^{1/2} - 1$$

In the weak field limit, $2M \ll r_o$ and r_e , so a first-order Taylor approximation gives us $z \approx M \left(\frac{1}{r_e} - \frac{1}{r_o} \right)$.

- In the previous example, if $r_e < r_o$, then $z > 0$, so as photons climb out of gravitational potential, they lose energy and are redshifted. Similarly, in the opposite case $r_o < r_e$, photons gain energy and are blueshifted as they follow the gravitational potential well.

3.9 Horizon and black holes

- The Schwarzschild metric is singular at $r = 2M$ or $r = \frac{2GM}{c^2}$. For the sun, $r = 2M_\odot \sim 3$ km, so the singularity is well within the star where the solution is not vacuum anymore. But when a star is massive enough, then the gravitational collapse when the star runs out of fuel can lead to objects for which this singularity becomes relevant.
- The Schwarzschild coordinates are valid from $2M$ to ∞ . To cover $r < 2M$, one has to find different coordinates that cover all of spacetime.

4 Generalizing spacetime geometry

The notions discussed in the second section transfer quite straightforwardly to the general formalism of differential geometry.⁴ In sections 2 and 3, we study GR in a *Riemannian geometry* $(\mathcal{M}, g)$, where $\mathcal{M}$ is the spacetime manifold and g is the metric. Because we have a notion of length and distance, it is possible to talk about its geodesics, magnitudes of vectors, areas, volumes, and so on. Given a metric, one can compute its Christoffel symbols, AKA its Levi-Civita connection. Thus, a Riemannian manifold comes naturally equipped with a covariant derivative. Recall that we stated that in GR, the covariant derivative satisfies a torsion-free condition. Another property of the Levi-Civita connection of the metric is that it is metric-compatible, a notion that we will discuss in this section. Therefore, Riemannian geometries are characterized by having curvature, but no torsion and no non-metricity. To define a covariant derivative on a manifold, no metric is needed in general: the covariant derivative is instead more generally induced by an affine connection Γ .⁵ The most general type of geometry is *metric-affine geometry*, consisting of the triple (M, g, Γ) . We subdivide metric-affine geometries by their curvatures, torsions, and non-metricities, which are notions which we define in the subsection below.

We add as a note that the classification of vectors as spacelike/timelike/null can be extended to a curve, which is spacelike/null/timelike if its tangent vector is everywhere spacelike/null/timelike, and to a hypersurface, which is spacelike/null/timelike if its normal vector is everywhere time-like/null/spacelike.

4.1 Curvature, torsion, and non-metricity

- As mentioned in the preceding footnote, the definition of the curvature tensor is the same in metric-affine geometry as in Riemannian geometry.
- Recall that curvature can be visualized as a measurement of the change of direction for a vector that is parallel transported around a closed loop. The notion of *torsion* can also be visualized in terms of parallel transport. Consider $A^\mu, B^\mu \in T_p\mathcal{M}$ defined at a point $p \in \mathcal{M}$, and consider parallel transporting A^μ along B^μ to its tip, and conversely parallel transporting B^μ along A^μ to its tip. In Euclidean geometry, this looks like a parallelogram: the tips of A^μ (resp. B^μ) after parallel transporting along its counterpart coincide. However, when a generic connection is

⁴I generally skip the details of differential geometry here.

⁵Recall that an affine connection Γ is defined with respect to a basis $\{e_\mu\}$ of $T\mathcal{M}$ by $\nabla_{e_\mu} e_\nu = \Gamma_{\mu\nu}^\alpha e_\alpha$, where ∇ is the covariant derivative satisfying the axioms stated in section 2; then, we can read off the component expression for the covariant derivative of a vector field A , which is $\nabla_\mu A^\nu = \partial_\mu A^\nu + \Gamma_{\mu\lambda}^\nu A^\lambda$. This expression should look familiar from subsection 2.2. Indeed, the expressions in 2.2 and 2.3 transfer perfectly to the case of affine geometry, where the Levi-Civita connection is replaced with an arbitrary affine connection.

being used to define parallel transport, the same operation described above (at an infinitesimal level) no longer leads to a closed parallelogram. Rather, the two parallel transports end on different points; the difference between the vectors describing their positions relative to p is measured by the torsion tensor. Concretely, defining C'^μ to be the vector from p to the tip of A^μ after parallel transporting along B^μ and defining the other vector C^μ in the obvious similar way, $C'^\alpha - C^\alpha = (\Gamma_{\mu\nu}^\alpha - \Gamma_{\nu\mu}^\alpha)\delta B^\mu \delta A^\nu$, so we define the *torsion tensor* to be the anti-symmetric part of the connection

$$T_{\mu\nu}^\alpha = \Gamma_{\mu\nu}^\alpha - \Gamma_{\nu\mu}^\alpha$$

- Consider a curve γ and let B^μ be the tangent vector to γ . If $A \in T\mathcal{M}$, the parallel transport equation $B^\alpha \nabla_\alpha A^\mu = 0$ allows us to determine how the magnitude changes under parallel transport. Recall that the magnitude is defined by the metric as $\|A\|^2 = g(A, A) = g_{\mu\nu} A^\mu A^\nu$. By Leibniz's rule, $B^\alpha \nabla_\alpha (g_{\mu\nu} A^\mu A^\nu) = (B^\alpha \nabla_\alpha g_{\mu\nu}) A^\mu A^\nu + 2g_{\mu\nu} (B^\alpha \nabla_\alpha A^\mu) A^\nu$. The last term vanishes due to the parallel transport equation, and we introduce the *non-metricity tensor*

$$Q_{\alpha\mu\nu} = \nabla_\alpha g_{\mu\nu}$$

so that the change $B^\alpha \nabla_\alpha (g_{\mu\nu} A^\mu A^\nu) = Q_{\alpha\mu\nu} B^\alpha A^\mu A^\nu$. The condition $Q_{\alpha\mu\nu} = 0$ is called the *metricity condition* and a connection which satisfies it is called *metric-compatible*.

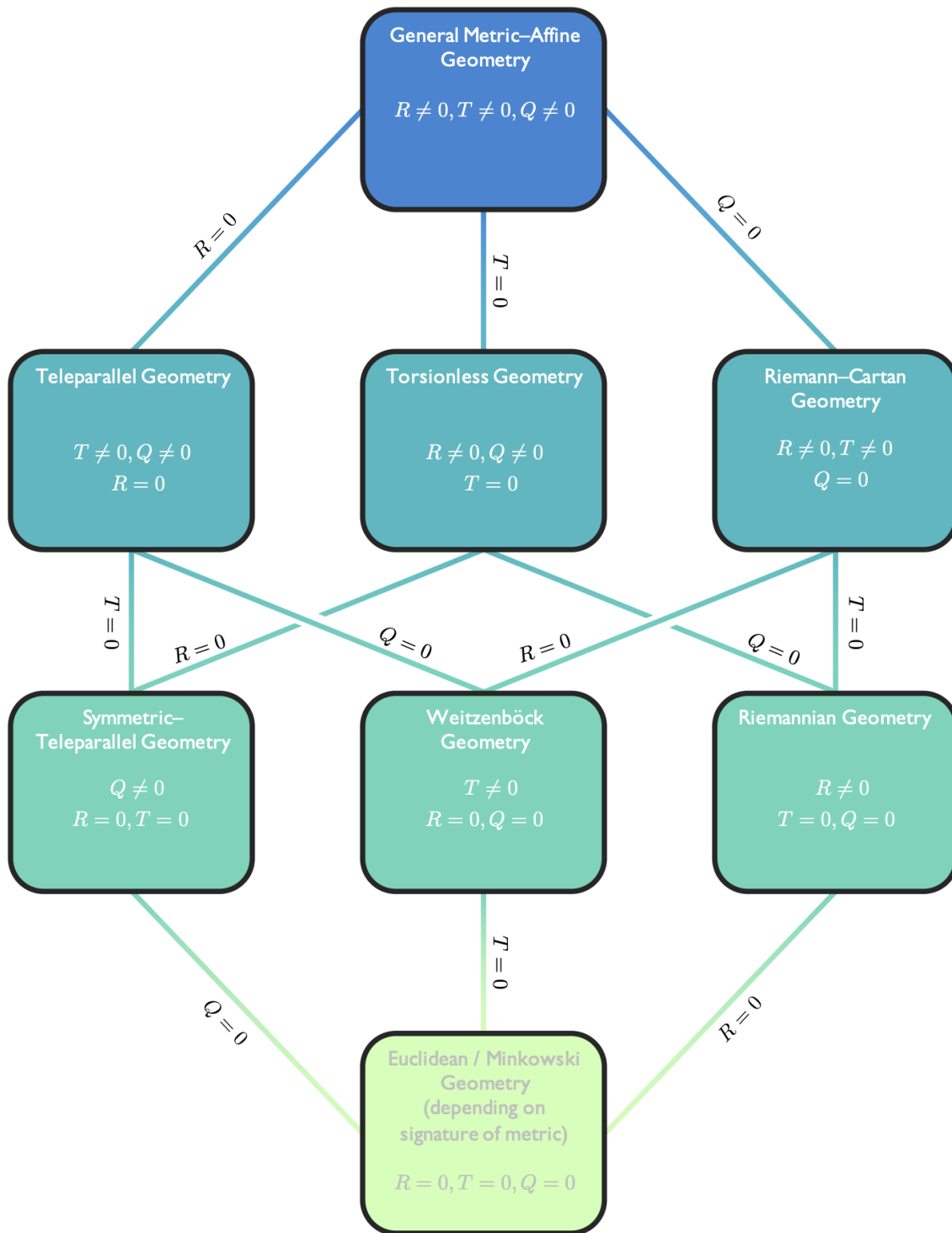
- Since the non-metricity tensor has three indices, two of which are symmetric, we can define two independent traces $Q_\alpha = Q_{\alpha\mu}^\mu$ and $\bar{Q}_\alpha = Q_{\mu\alpha}^\mu$.

4.2 Metric-affine geometry

As mentioned in the introduction of this section, we can subdivide metric-affine geometry by whether R, T, Q individually vanish. When all vanish, we have a flat spacetime.

5 Dynamics of the homogeneous universe

- The *cosmological principle* is that we should not assume that we are the center of the universe.
- The universe should appear isotropic to any free-falling observer throughout the universe. If the universe is isotropic around all points, then it is also homogeneous. To a first approximation, we will therefore treat the universe as perfectly homogeneous and isotropic.
- Spatial homogeneity and isotropy mean that the universe can be represented by a time-ordered sequence of three-dimensional spatial slices, each of which is homogeneous and isotropic. Such spaces are the maximally symmetric 3-spaces. Homogeneous and isotropic 3-spaces have constant curvature, which must be zero, positive, or negative. A 3-space with constant positive curvature can be represented as a 3-sphere S^3 embedded in four-dimensional Euclidean space E^4 , and a 3-space with constant negative curvature can be represented as a hyperboloid H^3 embedded in four-dimensional Lorentzian space $\mathbb{R}^{1,3}$.
- After a rescaling of coordinates $x^i \rightarrow ax^i$, where a is the radius of the 3-sphere in the spherical case and a^2 is an arbitrary constant in the arbitrary constant, the line element of a general



maximally symmetric 3-space can be written

$$d\ell^2 = a^2 \gamma_{ij} dx^i dx^j$$

$$\text{where } \gamma_{ij} = \delta_{ij} + k \frac{x_i x_j}{1 - k(x_k x^k)}, \text{ for } k = \begin{cases} 0 & \text{Euclidean} \\ +1 & \text{spherical} \\ -1 & \text{hyperbolic} \end{cases}$$

Notice we must take $a^2 > 0$ in order to have $d\ell^2$ positive at $x^i = 0$, and hence everywhere.

- The form of the spatial metric γ_{ij} depends on the choice of coordinates. In spherical polar coordinates, the metric becomes diagonal:

$$d\ell^2 = a^2 \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right]$$

where $d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2$. We may also define a radial coordinate $d\chi = dr/\sqrt{1 - kr^2}$ so that

$$d\ell^2 = a^2 [d\chi^2 + S_k^2(\chi) d\Omega^2] \text{ where } S_k(\chi) = \begin{cases} \sinh \chi & k = -1 \\ \chi & k = 0 \\ \sin \chi & k = +1 \end{cases}$$

- The *(Friedmann-)Robertson-Walker/FRW metric* for an expanding universe is obtained by including $d\ell^2$ as defined above into the spacetime line element and let the parameter a be an arbitrary function of time:

$$ds^2 = dt^2 - a^2(t) \gamma_{ij} dx^i dx^j$$

- Homogeneity and isotropy therefore reduce the ten independent coordinates of the spacetime metric into a *scale factor* $a(t)$ and a curvature parameter k .
- The coordinates x^i are called *comoving coordinates*, and the coordinates $x^i_{\text{phys}} = a(t)x^i$ are the *physical coordinates*. The physical velocity of an object is $v^i_{\text{phys}} = \frac{dx^i_{\text{phys}}}{dt} = a(t) \frac{dx^i}{dt} + \frac{da}{dt} x^i = v^i_{\text{pec}} + H x^i_{\text{phys}}$ where we have defined the *peculiar velocity* $v^i_{\text{pec}} = a(t) \dot{x}^i$ and the *Hubble parameter* $H = \frac{\dot{a}}{a}$. The $H x^i_{\text{phys}}$ term is called the *Hubble flow*. The peculiar velocity of an object is the velocity measured by a comoving observer, i.e. an observer who follows the Hubble flow.
- The FRW metric in polar coordinates reads

$$ds^2 = dt^2 - a^2(t) \left[\frac{dr^2}{1 - kr^2} + r^2 d\Omega^2 \right]$$

The line element has a rescaling symmetry $a \rightarrow \lambda a$, $r \rightarrow r/\lambda$, $k \rightarrow \lambda^2 k$. We then set $a_0 = a(t_0) = 1$.

- We can also write $ds^2 = dt^2 - a^2(t) [d\chi^2 + S_k^2(\chi) d\Omega^2]$; defining *conformal time* $d\tau = \frac{dt}{a(t)}$, this becomes $ds^2 = a^2(\tau) [d\tau^2 - (d\chi^2 + S_k^2(\chi) d\Omega^2)]$, which is a static Minkowski metric multiplied by a time-dependent conformal factor $a(\tau)$. Since light travels along null geodesics, the propagation of light in FRW is the same as in Minkowski space if we first transform to conformal time. Along the path, the change in conformal time equals the change in comoving distance.

- In this metric, the redshift for a photon emitted by a distant galaxy at time t_1 and observed on earth today is $z = \frac{a(t_0)}{a(t_1)} - 1$, where we recall that $a(t_0)$ is often defined to be 1.
- For nearby sources, we may expand $a(t_1)$ in a power series $a(t_1) = a(t_0)[1 + (t_1 - t_0)H_0 + \dots]$ where H_0 is the *Hubble constant* $H_0 = \frac{\dot{a}(t_0)}{a(t_0)}$. For close objects, $t_0 - t_1$ is simply the physical distance d (in units with $c = 1$). This gives us *Hubble's law* $z \approx H_0 d$. The slope in a redshift-distance diagram therefore measures the current expansion rate of the universe H_0 .
- Write the FRW metric $ds^2 = dt^2 - a^2(t)[d\chi^2 + S_k^2(\chi)d\Omega^2]$ where we have used the rescaling symmetry $a \rightarrow \lambda a$, $\chi \rightarrow \chi/\lambda$, $S_k^2 \rightarrow S_k^2/\lambda$ after choosing the scale factor to be dimensionless ($a(t_0) = 1$) to rewrite

$$S_k(\chi) = \begin{cases} R_0 \sinh(\chi/R_0) & k = -1 \\ \chi & k = 0 \\ R_0 \sin(\chi/R_0) & k = +1 \end{cases}$$

We use this to define the *metric distance* $d_m = S_k(\chi)$. In a flat universe, the metric distance equals to *comoving distance* χ ; the comoving distance between us and a galaxy at redshift z is $\chi(z) = \int_{t_1}^{t_0} \frac{dt}{a(t)} = \int_0^z \frac{dz}{H(z)}$, where the redshift evolution of the Hubble parameter depends on the matter content of the universe.

- Recall that absolute luminosity L is energy emitted per second and flux F is energy per second per receiving area. Type IA supernovae are called standard candles because they are believed to be objects of known absolute luminosity. The observed flux from a supernova explosion can be used to infer its (luminosity) distance. For a source at a fixed comoving distance χ in a static Euclidean space, $F = \frac{L}{4\pi\chi^2}$. In an FRW spacetime, this result is modified; at the time t_0 that the light reaches earth, the proper area of a sphere drawn around the supernova passing through Earth is $4\pi d_m^2$, so the fraction of light received in a telescope of aperture A is $A/4\pi d_m^2$. Moreover, the rate of arrival of photons is lower than the rate at which they are emitted, and the energy of the photons when they are received is less than the energy with which they are emitted. Hence in FRW spacetime

$$F = \frac{L}{4\pi d_m^2 (1+z)^2} = \frac{L}{4\pi d_L^2}$$

where we have defined the *luminosity distance* $d_L = d_m(1+z)$.

- Standard rulers are objects of known physical size D . Assuming that the object is at a comoving distance χ and the photons which we observe today were emitted at time t_1 , one could naively use for the distance between the observer and the emitter the *angular diameter distance* $d_A = \frac{D}{\delta\theta}$, where $\delta\theta$ is its angular size; the FRW metric implies that $D = a(t_1)S_k(\chi)\delta\theta = \frac{d_m}{1+z}\delta\theta$. We see that $d_A = \frac{d_L}{(1+z)^2}$.
- Recall the covariant conservation equation $\nabla_\mu T_\nu^\mu = 0$. The evolution of energy density is determined by the $\nu = 0$ equation. In an isotropic universe $T_0^i = 0$, so this reduces to $\dot{\rho} + \Gamma_{\mu 0}^\mu \rho - \Gamma_{\mu 0}^\lambda T_\lambda^\mu = 0$, which in the FRW metric reads $\dot{\rho} + 3\frac{\dot{a}}{a}(\rho + P) = 0$.
- The universe is filled with many different things, which we can classify by their contribution to pressure.

- *Matter* will be used to refer to all forms of matter for which pressure is much smaller than the energy density; in this case $\rho \propto a^{-3}$. This dilution of the energy density reflects the expansion of the volume $V \propto a^3$.
 - * Cosmologists refer to ordinary matter (nuclei and electrons) as *baryons*. Disgraceful.
 - * Most of the matter, however, is *dark matter*. We haven't figured out that part yet.
- *Radiation* is anything for which pressure is about a third of the energy density, $P = \frac{1}{3}\rho$. This is the case for a gas of relativistic particles, for which the energy density is dominated by the kinetic energy. In this case $\rho \propto a^{-4}$. The dilution now includes the redshifting of energy $E \propto a^{-1}$.
 - * *Photons* dominated the early universe. We detect them in those photons in the form of the CMB.
 - * *Neutrinos* behaved like radiation for most of the history of the universe, but recently their small masses have become relevant and they're starting to behave like matter.
 - * *Gravitons* might exist.
- The universe today seems to be dominated by a mysterious negative pressure component $P = -\rho$, which we call *dark energy*. In particular, the energy density of dark energy is constant; since the energy density doesn't dilute, energy has to be created as the universe expands. Note that this does not violate the conservation of energy in the gravitational context, since the proper conservation equation is the continuity equation above. Quantum field theory predicts this effect in the form of *vacuum energy*; the ground state energy of the vacuum corresponds to the stress-energy tensor $T^{\text{vac}} = \rho_{\text{vac}}g_{\mu\nu}$. Unfortunately, the predicted size of ρ_{vac} is greater than the observed ρ by a factor of 10^{120} .
- We note before moving on that the left-hand side of the Einstein equation $G_{\mu\nu} = 8\pi GT_{\mu\nu}$ isn't uniquely defined, and that we can add the term $-\Lambda g_{\mu\nu}$ for some *cosmological constant* Λ without violating $\nabla^\mu T_{\mu\nu} = 0$. Modern practice is to move this term to the right hand side and treat it as a contribution to the stress-energy tensor of the form $T_{\mu\nu}^{(\Lambda)} = \frac{\Lambda}{8\pi G}g_{\mu\nu} = \rho_\Lambda g_{\mu\nu}$. This is the same form as the stress-energy tensor from vacuum energy.
- In the FRW metric, the non-vanishing components of the Ricci tensor are

$$R_{00} = -3\frac{\ddot{a}}{a},$$

$$R_{ij} = \left[\frac{\ddot{a}}{a} + 2\left(\frac{\dot{a}}{a}\right)^2 + 2\frac{k}{a^2} \right] g_{ij}$$

and the Ricci scalar is

$$R = -6 \left[\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} \right]$$

The nonzero components of the Einstein tensor are

$$G_0^0 = 3 \left[\left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} \right]$$

$$G_j^i = \left[2\frac{\ddot{a}}{a} + \left(\frac{\dot{a}}{a}\right)^2 + \frac{k}{a^2} \right] \delta_j^i$$

- Recall that the stress-energy tensor for a perfect fluid in a rest frame is $T^\mu_\nu = \text{diag}(\rho, -P, -P, -P)$ (where the signs on the P simply come from our change in metric signature). Using this, we get the *Friedmann equations*

$$\begin{aligned}\left(\frac{\dot{a}}{a}\right)^2 &= \frac{8\pi G}{3}\rho - \frac{k}{a^2} \\ \frac{\ddot{a}}{a} &= -\frac{4\pi G}{3}(\rho + 3P)\end{aligned}$$

where ρ and P are understood as the sum of all contributions to the energy density and pressure in the universe. The first Friedmann equation can also be written in terms of the Hubble parameter as

$$H^2 = \frac{8\pi G}{3}\rho - \frac{k}{a^2}$$

- Use superscripts 0 to denote quantities evaluated today at $t = t_0$. A flat universe $k = 0$ corresponds to the *critical density* today $\rho_{\text{crit},0} = \frac{3H_0^2}{8\pi G}$, which we use to define dimensionless density parameters $\Omega_{I,0} = \frac{\rho_{I,0}}{\rho_{\text{crit},0}}$.
- We write ρ_r for contribution from radiation (ρ_γ for photons, ρ_ν for neutrinos), ρ_m for matter (ρ_c for cold dark matter, ρ_b for baryons), and ρ_Γ for vacuum energy.
- The first Friedmann equation can then be written

$$H^2(a) = H_0^2 \left[\Omega_{r,0} \left(\frac{a_0}{a}\right)^4 + \Omega_{m,0} \left(\frac{a_0}{a}\right)^3 + \Omega_{k,0} \left(\frac{a_0}{a}\right)^2 + \Omega_{\Gamma,0} \right]$$

where we have defined a curvature density parameter $\Omega_{k,0} = -k/(a_0 H_0)^2$. In literature, the subscript 0 is normally dropped (so it is assumed that we speak of today unless otherwise stated). We will also use $a_0 = 1$.

- Observations show that $|\Omega_k| \leq 0.01$, $\Omega_r = 9.4 \times 10^{-5}$, $\Omega_m = 0.32$, $\Omega_\Lambda = 0.68$. The equation of state of dark energy appears to be that of a cosmological constant $w_\Lambda \approx 1$. Matter splits into $\Omega_b = 0.05$ and $\Omega_c = 0.27$. Since curvature makes up less than 1% of the cosmic energy budget we set $\Omega_k = 0$ in what follows.
- The different scalings of radiation (a^{-4}), matter (a^{-3}), and vacuum energy (a^0) imply that for most of its history the universe was dominated by a single component (first radiation, then matter, then vacuum energy). We parametrize this component by its equation of state w_I . For a flat, single-component universe, the Friedmann equation reduces to

$$\frac{\dot{a}}{a} = H_0 \sqrt{\Omega_I} a^{-\frac{3}{2}(1+w_I)}$$

REMEMBER TO LATEX TABLE 1.1 BAUMANN PG 31

- At $a_{eq} = \Omega_r/\Omega_m \approx 3 \times 10^{-4}$, we had a two-component universe. In a flat universe filled with a mixture of matter and radiation, the Friedmann equations are

$$\begin{aligned}(a')^2 &= \frac{8\pi G}{3}\rho a^4 \\ a'' &= \frac{4\pi G}{3}(\rho - 3P)a^3\end{aligned}$$

where primes denote derivatives w.r.t. conformal time and $\rho = \rho_m + \rho_r = \frac{\rho_{eq}}{2} \left[\left(\frac{a_{eq}}{a} \right)^3 + \left(\frac{a_{eq}}{a} \right)^4 \right]$. Since $\rho_r - 3P_r = 0$ and $\rho_m a^3 = \frac{1}{2} \rho_{eq} a_{eq}^3$, it follows that $a'' = \frac{2\pi G}{3} \rho_{eq} a_{eq}^3$. Imposing $a(\tau = 0) = 0$ fixes one of the constants of integration, so the differential equation has the solution

$$a(\tau) = a_{eq} \left[\left(\frac{\tau}{\tau_\star} \right)^2 + 2 \left(\frac{\tau}{\tau_\star} \right) \right], \text{ where } \tau_\star = \left(\frac{\pi G}{3} \rho_{eq} a_{eq}^2 \right)^{-1/2} = \frac{\tau_{eq}}{\sqrt{2} - 1}$$

For $\tau \ll \tau_{eq}$, we recover the radiation-dominated limit $a \propto \tau$, while for $\tau \gg \tau_{eq}$ we get the matter-dominated limit $a \propto \tau^2$.

6 Cosmological inflation

- Standard cosmology predicts that the early universe was made of many causally disconnected regions of space. The *horizon problem* is to explain that why these apparently disjoint patches of space have very nearly the same densities and temperatures.
- Define the reduced Planck mass $M_{pl} = \sqrt{\frac{hc}{8\pi G}}$ so that the Friedmann equation is $H^2 = \rho/(3M_{pl}^2)$.
- In an isotropic spacetime we can always define the coordinate system so that light travels purely in the radial direction. The evolution is then determined by a two-dimensional line element $ds^2 = a^2(\tau)[d\tau^2 - d\chi^2]$ (where we have again used the reparametrization $d\chi = dr/\sqrt{1 - kr^2}$ to deal with curvature); photons travel along the null geodesics with paths $\Delta\chi(\tau) = \pm\Delta\tau$, where the plus sign corresponds to outgoing photons and the minus to incoming photons.
- The maximal comoving distance that light can travel between two times $\tau_1, \tau_2 > \tau_1$ is simply $\tau_2 - \tau_1$; if the Big Bang starts with a singularity at $t_i = 0$ (note the Big Bang singularity is a moment in time but not space), then the greatest comoving distance from which an observer at time t will be able to receive signals travelling at the speed of light is given by $\chi_{ph}(\tau) = \tau - \tau_i = \int_{\tau_i}^{\tau} \frac{d\tau}{a(\tau)}$. This is the *(comoving) particle horizon*. The size of the particle horizon at τ may be visualized by the intersection of the past light cone of an observer p with the spacelike hypersurface $\tau = \tau_i$. Causal influences have to come from within this region. Only comoving particles whose worldlines intersect that past light cone of p can send a signal to an observer at p ; the boundary of the region containing such worldlines is the particle horizon at p .
- In comoving coordinates, the greatest distance from which an observer at time t_f will receive signals emitted at any time later than t is given by $\chi_{eh}(\tau) = \tau_f - \tau = \int_{\tau}^{\tau_f} \frac{d\tau}{a(\tau)}$. This is called the *(comoving) event horizon*. τ_f denotes the final moment of (conformal) time. This may be finite even if physical time is infinite; whether or not this is the case depends on the form of $a(t)$. τ_f is finite for our universe if dark energy is really a cosmological constant.
- We can write $\chi_{ph}(\tau) = \int_{a_i}^a \frac{da}{aH} = \int_{\ln a_i}^{\ln a} (aH)^{-1} d\ln a$ where $a_i = 0$ corresponds to the Big Bang singularity. We call $(aH)^{-1}$ the *comoving Hubble radius*. For a universe dominated by a fluid with constant equation of state $w = P/\rho$, we have $(aH)^{-1} = H_0^{-1} a^{\frac{1}{2}(1+3w)}$. All familiar matter sources satisfy the strong energy condition $1 + 3w > 0$, so we used to assume that the comoving Hubble radius increases as the universe expands, in which case the integral is dominated by the upper limit and receives vanishing contributions from early times, which is explicit in the example

of a perfect fluid, in which case $\chi_{ph}(a) = \frac{2H_0^{-1}}{(1+3w)} \left[a^{\frac{1}{2}(1+3w)} - a_i^{\frac{1}{2}(1+3w)} \right] = \tau - \tau_i$. Defining $\tau_i = \frac{2H_0^{-1}}{(1+3w)} a_i^{\frac{1}{2}(1+3w)} \xrightarrow{a_i \rightarrow 0, w > -\frac{1}{3}} 0$, the comoving horizon is finite, $\chi_{ph}(t) = \frac{2}{(1+3w)} (aH)^{-1}$. So in standard cosmology, $\chi_{ph} \sim (aH)^{-1}$.

- Conjecture a phase of decreasing Hubble radius in the universe. This requires a strong-energy-condition-violating fluid. The integral above is dominated by the lower limit; the Big Bang singularity is pushed to negative conformal time, $\tau_i \rightarrow -\infty$. The widely separated points in the CMB now have enough time to intersect before the time τ_i . In inflationary cosmology, $\tau = 0$ isn't the initial singularity, but instead becomes only a transition point between inflation and the standard Big Bang evolution; there is time both before and after $\tau = 0$.
- From the relation $\frac{d}{dt}(aH)^{-1} = \frac{d}{dt}(\dot{a})^{-1} = -\frac{\ddot{a}}{(\dot{a})^2}$, we see that a shrinking comoving Hubble radius implies accelerated expansion $\ddot{a} > 0$. As such, inflation is often defined as a period of acceleration.
- Alternatively, we may write $\frac{d}{dt}(aH)^{-1} = -\frac{\dot{a}H + a\dot{H}}{(aH)^2} = -\frac{1}{a}(1 - \varepsilon)$ where $\varepsilon = -\frac{\dot{H}}{H^2}$; the shrinking Hubble sphere therefore also corresponds to $\varepsilon < 1$.
- For perfect inflation $\varepsilon = 0$, the spacetime becomes *de Sitter space* $ds^2 = dt^2 - e^{2Ht}dx^2$ where $H = \partial_t \ln a = \text{const}$. Inflation has to end, so it shouldn't correspond to perfect de Sitter space; for small but finite $\varepsilon \neq 0$, this line element is still a good approximation to the inflationary background, so we sometimes refer to inflation as a quasi-de Sitter period.
- For a perfect fluid with pressure P and density ρ , the Friedmann equation $H^2 = \rho/(3M_{pl}^2)$ and continuity together imply $\varepsilon = \frac{3}{2} \left(1 + \frac{P}{\rho} \right) < 1 \iff w = \frac{P}{\rho} < -\frac{1}{3}$, i.e. that inflation requires negative pressure or a violation of the SEC. Moreover, $\left| \frac{d \ln \rho}{d \ln a} \right| = 2\varepsilon < 1$, so for small ε the energy density is nearly constant.
- Therefore, a given FRW spacetime with time-dependent $H(t)$ corresponds to cosmic acceleration iff $\varepsilon = -\frac{\dot{H}}{H^2} = -\frac{d \ln H}{dN} < 1$, where $dN = Hdt$ measures the number of e -folds N of inflationary expansion. Solving the horizon problem requires N to be at least 40 to 60 e -folds, requiring ε to be small for long enough; this condition is measured by $\eta = \frac{d \ln \varepsilon}{dN} = \frac{\dot{\varepsilon}}{H\varepsilon}$, since for $|\eta| < 1$, the fractional change of ε per Hubble time is small and inflation persists.
- As a toy model for inflation consider the scalar field $\phi(t, \vec{x})$ which we call the *inflaton*. Consistency with the symmetries of FRW spacetime requires that the background value of the inflaton only depends on time. From the time component $T_0^0 = \rho_\phi$ of the stress-energy tensor, we infer that $\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi)$, and from the space-space component $T_j^i = -P_\phi \delta_j^i$, we find that the pressure is $P_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi)$. Therefore, a field configuration leads to inflation, $P_\phi < -\frac{1}{3}\rho_\phi$, if potential energy dominates over the kinetic energy.
- After substituting ρ_ϕ into the first Friedmann equation, we get

$$H^2 = \frac{1}{3M_{pl}^2} \left[\frac{1}{2}\dot{\phi}^2 + V \right]$$

And from the second Friedmann equation

$$\dot{H} = -\frac{1}{2} \frac{\dot{\phi}^2}{M_{pl}^2}$$

Combining this with the equation obtained from taking the time derivative of the first equation, we get the Klein-Gordon equation

$$\ddot{\phi} + 3H\dot{\phi} + V' = 0$$

The potential acts like a force, while the expansion of the universe adds friction.

- From the above we find $\varepsilon = \frac{\frac{1}{2}\dot{\phi}^2}{M_{pl}^2 H^2}$. Inflation therefore occurs if the kinetic energy makes only a small contribution to the total energy $\rho_\phi = 3M_{pl}^2 H^2$. This situation is called *slow-roll inflation*. For this condition the acceleration of the scalar field has to be small. Define $\delta = -\frac{\ddot{\phi}}{H\dot{\phi}}$. Then substituting things gives $\eta = 2(\varepsilon - \delta)$. Hence, $\{\varepsilon, |\delta|\} \ll 1 \implies \{\varepsilon, |\eta|\} \ll 1$.
- Using these conditions to simplify the equations of motion is called the *slow-roll approximation*. The condition $\varepsilon \ll 1$ implies $\frac{1}{2}\dot{\phi}^2 \ll V$, leading to the simplification of the Friedmann equation,

$$H^2 \approx \frac{V}{3M_{pl}^2}$$

In the slow-roll approximation, the Hubble expansion is determined completely by the potential energy; $|\delta| \ll 1$ simplifies the Klein-Gordon equation to

$$3H\dot{\phi} \approx -V'$$

From these, $\delta + \varepsilon \approx M_{pl}^2 \frac{V''}{V}$, hence we can assess whether a given potential $V(\phi)$ leads to slow-roll inflation is to compute the *potential slow-roll parameters*

$$\epsilon_v = \frac{M_{pl}^2}{2} \left(\frac{V'}{V} \right)^2, \quad |\eta_v| = M_{pl}^2 \frac{|V''|}{V}$$

Successful slow-roll inflation occurs when these parameters are small. In contrast, η, ε are called *Hubble slow-roll parameters*; during slow roll, $\epsilon_v \approx \varepsilon$ and $\eta_v \approx 2\varepsilon - \frac{1}{2}\eta$.

- The total number of e -folds of accelerated expansion are $N_{tot} = \int_{a_I}^{a_E} d \ln a = \int_{t_I}^{t_E} H(t) dt$, where t_I, t_E are defined as the times when $\varepsilon(t_I) = \varepsilon(t_E) = 1$. In the slow-roll regime, $H dt = \frac{1}{\sqrt{2\varepsilon}} \frac{|d\phi|}{M_{pl}} \approx \frac{1}{\sqrt{2\varepsilon_v}} \frac{|d\phi|}{M_{pl}}$, so $N_{tot} = \int_{\phi_I}^{\phi_E} \frac{1}{\sqrt{2\varepsilon_v}} \frac{|d\phi|}{M_{pl}}$ where ϕ_I, ϕ_E are the boundaries of the interval where $\epsilon_v < 1$. The largest scales observed in the CMB are produced about 60 e -folds before the end of inflation; $N_{cmb} = \int_{\phi_{cmb}}^{\phi_E} \frac{1}{\sqrt{2\varepsilon_v}} \frac{|d\phi|}{M_{pl}} \approx 60$. A successful solution to the horizon problem requires $N_{tot} > N_{cmb}$.
- During inflation, most of the energy density in the universe is in the form of the inflaton potential; inflation ends when the potential steepens and the inflaton field picks up kinetic energy. The energy in the inflaton sector is then transferred to the Standard Model particles. This process is called *reheating* and begins with the *Hot Big Bang*.

- After inflation, ϕ begins to oscillate at the bottom potential. Assuming $V(\phi) \approx \frac{1}{2}m^2\phi^2$ near the minimum of $V(\phi)$ where the amplitude of ϕ is small, the inflaton is still homogeneous, so its EOM is $\ddot{\phi} + 3H\dot{\phi} = -m^2\phi$. The expansion time scale becomes much longer than the oscillation period, $H^{-1} \gg m^{-1}$. We can then neglect the friction term and the field undergoes oscillations with frequency m . Writing continuity as $\dot{\rho}_\phi + 3H\rho_\phi = -3HP_\phi = -\frac{3}{2}H(m^2\phi^2 - \dot{\phi}^2)$, the r.h.s. averages to zero over one oscillation period, so the oscillating field therefore behaves like pressureless matter with $\rho_\phi \propto a^{-3}$. The fall in energy density is reflected in a decrease of the oscillation amplitude.
- The inflaton has to couple to SM fields, so the energy stored in it will then be transferred to ordinary particles. If the decay is slow, in which case the inflaton can only decay into fermions, the inflaton energy density follows $\dot{\rho}_\phi + 3H\rho_\phi = -\Gamma_\phi\rho_\phi$, where Γ_ϕ parameterizes inflaton decay rate. If the inflaton can decay into bosons, the decay may be very rapid, involving *parametric resonance* sourced by Bose condensation events. This kind of rapid decay is called *preheating*, since the bosons are far from thermal equilibrium.
- The particles produced by inflaton decay will interact and the resulting particle soup will eventually reach thermal equilibrium with some temperature T_{rh} . Reheating temperature is determined by the energy density ρ_{rh} at the end of the reheating epoch. $\rho_{rh} < \rho_{\phi,E}$ where $\rho_{\phi,E}$ is the inflaton energy density at the end of inflation; if reheating takes a long time, $\rho_{rh} \ll \rho_{\phi,E}$. Though it can be quite involved, the evolution of the gas particles into a thermal state is assumed to happen eventually; some particles that never reach thermal equilibrium (e.g. gravitinos) might exist, since their interactions are too weak. As long as the momenta of the particles are much higher than their masses, the energy density of the universe behaves like radiations regardless of the momentum space distribution. After thermalisation of at least the baryons, photons, and neutrinos is complete, the standard Hot Big Bang era begins.